

Best-fit lines, anomalous results and conclusions

Working scientifically
Both papers · Combined and Separate
10 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 4 starred (★) questions first.

	QUESTION	ANSWER	✓
1	What is a line of best fit?	One smooth line showing the overall trend of the data.	☐
2 ★	Describe how a line of best fit should be drawn.	As one thin line with points balanced above and below it.	☐
3	State what must not be done when drawing a line of best fit.	Joining the points dot-to-dot.	☐
4 ★	What is an anomalous result (an outlier)?	A result that does not fit the pattern of the others.	☐
5 ★	Describe what you should do when you find an anomalous result.	Check the recording and plotting, then repeat the measurement. If it is still anomalous, leave it out of the mean and the line of best fit.	☐
6	What is a trend?	The overall relationship shown by the data.	☐
7	State what a good conclusion must include.	The relationship between the variables, supported by values with units.	☐
8 ★	Describe the graph of two quantities that are directly proportional.	A straight line through the origin.	☐
9	State the three parts of a good evaluation point.	The problem, its effect on the data, and an improvement that fixes it.	☐
10	What is peer review?	When other scientists check the method, the evidence and the claims.	☐